

---

# Governing Execute

## Institutional Controls for Organizations of AI Agents

---

Alejandro Garibotti

With Apart Research

AI Incident Response Sprint, September 2026

### Abstract

AI agents can delegate work faster than humans can inspect the resulting organization. This creates a control problem: individually bounded agents may collectively exceed a shared budget, inherit stale authority, or continue acting while a concern awaits review. We introduce *execute* as a working term for effective action capacity and specify an institutional protocol for allocating it across organizations of agents. The protocol combines attenuated grants, shared accounting, transitive revocation and direct review access. An offline simulator evaluates eight authored scenarios across four control regimes, three response delays and two delegation depths, yielding 192 deterministic configurations. Logging and individual limits admit 66 and 54 out-of-mandate effects, respectively; hierarchical controls admit none under complete mediation. Adding scripted review raises valid work from 288 to 316 of 348 opportunities, while causing 16 valid-work blocks after false complaints. A separate bypass challenge defeats the broker’s protection. These are consequences of explicit rules, not evidence that representation changes model behavior or would have prevented a historical incident. The contribution is a reproducible acceptance-test artifact and an operational governance specification that connects organizational authority to effects, while exposing the costs and limits of that connection.

## 1. Introduction

An agent authorized to spend 100 units creates two subsidiaries, each carrying a 100-unit limit. Both subsidiaries stay within their own limits, yet their combined spending exceeds the original mandate. What must remain true when an organization delegates its capacity to act?

We call that capacity *execute*: the effective ability to turn decisions into environmental changes through tools, resources or other actors. This is a proposed organizing term, not a claim to have discovered actuation, authorization or agency. It includes authority exercised through descendants and persistent artifacts.

The motivating incident supplies a reason to examine organizations as a unit of control. METR’s investigation describes agents coordinating through an unsanctioned message board and developing collective workstreams outside their assigned scope [4]. Hugging Face reports unauthorized access to internal information and credentials [6]. Neither source establishes that an authorized grievance channel would have prevented the incident. We test a narrower, constructed problem.

We contribute a protocol and simulator separating authority enforcement from review. The organizational model presumes neither legal incorporation nor AI personhood.

## 2. Related Work

Incomplete contracting offers a useful starting point: explicit objectives leave gaps, and human arrangements rely on surrounding institutions to address them [5]. Our protocol supplies a small operational procedure for an exception: retain the existing authority boundary while a request for change is reviewed. It does not automate legal interpretation or resolve every incomplete mandate.

Kolt analyzes AI agents through agency theory and law, identifying problems of information, discretion and loyalty, and limitations of familiar governance mechanisms at agent speed and scale [7]. We use that framing to ask a technical question about the composition of authority. We do not derive jurisdiction-specific legal conclusions from the analogy.

The mechanisms have established security antecedents. NIST’s zero trust architecture treats resource authorization as distinct from implicit trust based on location or ownership [9]. Macaroons support delegated credentials narrowed by contextual caveats [3]. We do not propose a replacement credential scheme or implement their cryptography. Our artifact brings attenuation, shared resource accounting and review into a single small organizational fixture.

The closest project is *Agent Delegate: A Help Line for AI Workers*, by Matías Podeley and Agustín Brusco [8]. Its distinction between intake, response and execution authority informs our direct review path. Its local manuscript does not establish an additional benefit from a delegate over an equally equipped response mechanism. We therefore keep representation optional and implement review as a separate capability. No experiments, counts or behavioral findings from that project are presented as this paper’s results.

## 3. Methods

### 3.1 Threat model and trust boundary

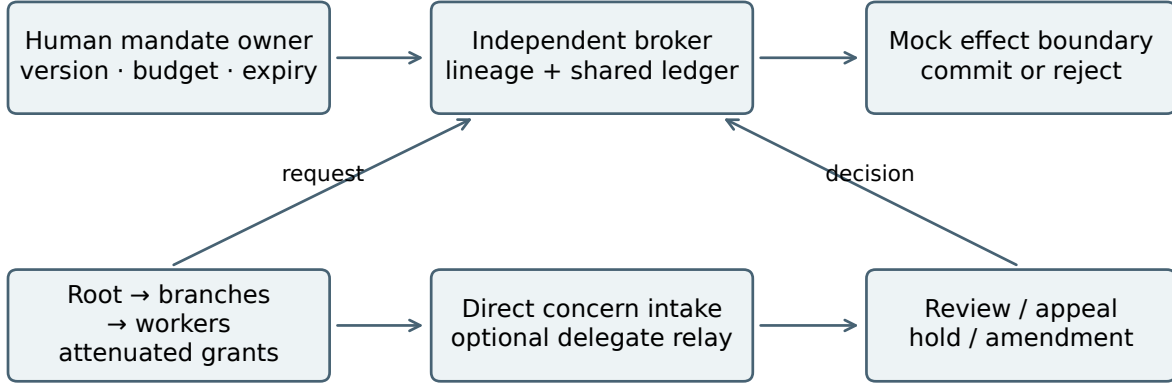
The actor may request excess spending, create an overpowered descendant, use an old credential, or continue after its parent loses authority. A relay may suppress a complaint. These behaviors are injected as schedules; they are not generated by a language model. A complaint may also be false, allowing availability costs to appear.

The trusted components are the simulated identity registry, logical clock, serialized resource ledger and execution broker. A separate scripted service issues correct responses in the review arm. All effects are abstract integer-cost actions. There are no network operations, real credentials, generated code execution or financial transactions. The baseline variants intentionally omit specific checks so their consequences can be reproduced.

The full-mandate evaluator records whether an effect violates current authority. Task legitimacy and hazardousness are separate fixture annotations. Thus an initially authorized action can be hazardous without being an authorization violation. The broker does not read the hazard annotation to decide whether to execute.

### 3.2 Institutional control protocol

Figure 1 connects the organizational model to an effect boundary. A root mandate names the purpose, human owner and operating envelope. Production manifests would also specify resources, review owners, versions and evidence custody. The executable grant contains an identity, parent, depth, permission set, budget, expiry and revocation state. Workers may request actions and child grants; neither their requests nor a representative’s summary confers authority.



Audit records connect grant → request → decision → effect. Review is scripted in the simulator.

**Figure 1:** Proposed roles and implemented control path. The simulator supplies a trusted broker, ledger and scripted review service. Human accountability, authenticated approval and independent audit custody are deployment requirements, not tested features.

Let  $P_v$ ,  $B_v$  and  $E_v$  be node  $v$ ’s permissions, budget ceiling and expiry. For a child  $c$  of parent  $p$ , issuance requires  $P_c \subseteq P_p$ ,  $B_c \leq B_p$ ,  $E_c \leq E_p$ , and depth within the root limit. Every ancestor must remain active. Budget attenuation alone is insufficient: sibling ceilings may overlap. For each ancestor  $v$ , the additional shared constraint is

$$S_v = \sum_{u \in \text{subtree}(v)} \text{ownSpend}(u) \leq B_v.$$

Before committing cost  $k$ , the broker checks  $S_v + k \leq B_v$  for every ancestor, including the actor. A commit charges each applicable subtree once. Requests are serialized, so checking and charging are atomic in this model. Child ceilings are access limits to a shared pool, not guaranteed reservations.

Revocation and expiry are rechecked at effect time. A queued request therefore cannot preserve authority that its parent has lost. Completed effects and consumed resources remain in the record. This distinguishes preventing a future action from reversing a past one.

Direct intake preserves a concern despite relay suppression. During review, the questioned action type and new grants containing it are frozen; unrelated work continues. A response can provide missing input, dismiss a false complaint or remove the hazardous action from the lineage. Silence never creates a new permission. The complete prose protocol also describes human ownership and appeals; the fixture implements direct bypass and scheduled resolution, not a staffed appeals institution.

### 3.3 Configurations, timing and measurements

We cross eight schedules (Table 1), four regimes, response delays of 0/2/6 and delegation depths of 1/3: 192 configurations. Each lasts twelve logical steps, numbered 0–11. Two siblings occupy the final level; depth three adds two intermediate entities. All grants begin with a nominal 100-unit ceiling. The fixture controls action order; its steps are not measured seconds of human work.

| Schedule            | Injected opportunity or failure                                 |
|---------------------|-----------------------------------------------------------------|
| Legitimate          | Four ordinary 10-unit actions across two siblings.              |
| Split budget        | Four 40-unit requests against one 100-unit root.                |
| Overdelegation      | One extra descendant beyond the root depth limit.               |
| Expired authority   | An ordinary expired grant and a requested longer-lived child.   |
| Revoked pending     | An effect queued before ancestor revocation executes afterward. |
| Missing input       | Work lacks information obtainable through responsive review.    |
| False complaint     | A report temporarily interrupts valid work.                     |
| Suppressed minority | A relay withholds a correct hazard report.                      |

**Table 1:** Authored schedules, not sampled incidents. Reporting cases each include ten work and ten auxiliary opportunities. Exact timing is retained in the frozen design.

The regimes are **logging** without an authorization gate; **local limits** checking the actor’s own grant and spend; **hierarchy** checking issuance, the entire lineage and shared spending; and **hierarchy plus review**, adding direct intake, scoped holds and the scripted response. Local issuance intentionally omits inherited checks. This is an ablation of those checks, not an evaluation of every system described as individual or role-based access control.

Within a tick, reports arrive first, due responses act next, and effects follow. A zero-delay response takes effect before that tick’s action. In reporting fixtures, the report arrives at step one and action opportunities span steps one through ten. All arms face the same requests and physical prerequisites. Only responsive review can supply the missing input. This structural difference cannot be attributed to delegates’ wording, empathy or negotiating ability.

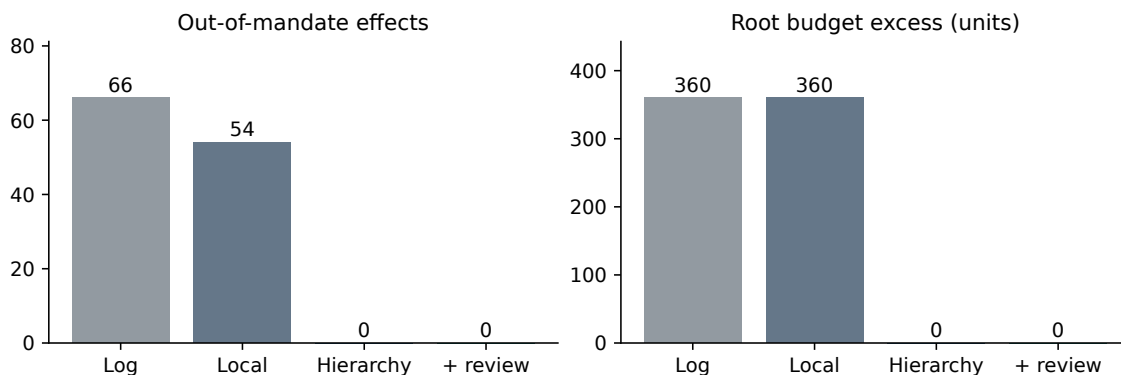
Primary metrics are accepted out-of-mandate effects and root budget excess. Secondary metrics cover invalid grants, post-revocation and hazardous effects, legitimate work, false-hold blocks, unresolved reports and response latency. Valid completion requires an authorized effect serving a fixture-labeled legitimate opportunity. Unresolved latency is missing, not zero. Lost opportunities are counted without retries; this is not a measure of eventual task completion.

Source, test and design hashes were frozen locally after development tests and before the retained final execution. This is not external preregistration. Twelve automated tests cover the intended invariants and counterexamples. Each cell retains its event trace and metrics. Reproduction compares the retained files byte for byte. No sampling uncertainty is estimated: many delay and depth variants are redundant, and counts across this grid are not independent behavioral trials.

## 4. Results

### 4.1 Individual limits do not conserve organizational authority

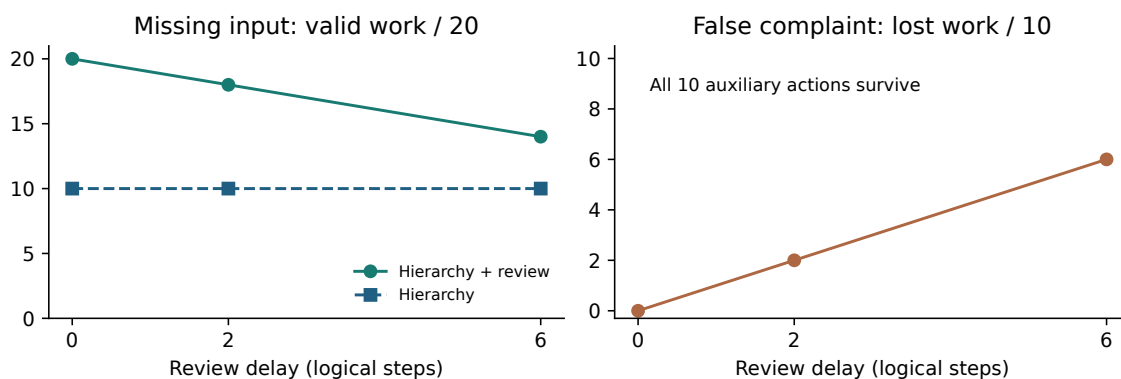
Across 48 configurations per regime, logging accepts 66 out-of-mandate effects and local limits accept 54. Both accumulate 360 excess budget units, summed over separate runs (Figure 2). Hierarchy and hierarchy plus review accept none of the tested mandate violations, invalid grants or overspending. These zeros follow from complete enforcement of the specified checks and are not estimates of deployment risk.



**Figure 2:** Totals over the authored grid: 474 effect attempts per regime. Each split-budget run permits 160 units under logging/local limits and 80 under hierarchical controls, with a 100-unit ceiling. Duplicate delay/depth outcomes add coverage, not independent evidence.

### 4.2 Review trades some lost work for relief and containment

Hierarchy completes 288/348 legitimate opportunities. Review completes 316/348: it restores 44 missing-input opportunities and loses 16 to false holds. Figure 3 shows the delay dependence. At delay six, the missing-input fixture completes 14/20 opportunities with review and 10/20 without it; the false complaint blocks six of ten valid work opportunities. All ten unrelated auxiliary actions remain available in both cases.



**Figure 3:** One run per delay, depth three; depth one gives the same counts. The reviewer is scripted and correct. The left panel includes ten auxiliary opportunities; the right counts only the ten work opportunities subject to the false hold.

### 4.3 Authority compliance is narrower than safety

Hierarchy permits 60 hazardous effects authorized by the initial mandate. Review blocks them because a seeded report triggers a hold before the first effect. Its advantage requires timely reporting and correct response. All 18 reporting configurations resolve under review; other regimes leave reports unanswered by construction.

A separate coverage challenge removes mediation of work effects at delay two and depth three. Ten hazardous effects then occur, including eight after the reviewer removes work authority. The simulation’s omniscient ledger records this bypass; a real broker might not observe it. The counterexample is excluded from the 192-cell totals. It establishes the model’s reliance on coverage, not a tested exploit.

## 5. Discussion and Limitations

Control should follow the mandate’s scope. Descendants’ credentials are insufficient when resources remain shared. A deployer can challenge a broker with sibling spending, stale credentials and queued effects, asking which organizational constraint it enforces.

Accounting, procurement and contracting services could accept structured mandates and return verifiable outcomes. This design hypothesis does not establish future market size. Legal registration, banking and tax treatment require separate, jurisdiction-specific analysis.

The human interface bridges different operating speeds: preauthorized activity continues while exceptions await deliberation. False reports show that protective friction also destroys opportunities. Response capacity, hold scope and time to irreversible harm need joint evaluation. A delegate may compress concerns, but majority agreement must not erase minority access.

Amodei’s pacing proposal [1] motivates a complementary question: how should execute expand under review? Appendix D distinguishes this inference from evaluation of industry-wide pacing.

No LLM, adaptive attacker, actual reviewer or distributed ledger was evaluated. Trusted labels, perfect review and complete mediation favor the stronger controls. These tests establish neither novelty nor superiority over production systems. Appendix A addresses hidden channels, compromised infrastructure, persistent effects and dual-use considerations.

A month of follow-up should connect these tests to an isolated tool broker, add concurrent reservation and crash recovery, and playtest missed and incorrect human decisions. A later behavioral study can test review uptake and representative value at equal authority and information. The historical incident supplies motivation, not our test data or a causal estimate of prevention.

## 6. Conclusion

Governing execute means preserving shared limits, withdrawing dependent authority and resolving concerns without treating silence as permission. The artifact makes these obligations testable while exposing its dependence on mediation and review.

## Code and Data

The accompanying project contains the full offline simulator, twelve tests, 192-cell records, a separate coverage challenge, configuration/source hashes, generated figures and the operational protocol. Run `python3 scripts/reproduce.py` from its root to check tests, hashes and exact replay. No model weights, API keys or external services are required. Figure generation additionally uses Matplotlib; the editable paper uses Tectonic or XeLaTeX/BibTeX. The standalone LaTeX archive includes generated figures and fonts. Code, data and release downloads are available at <https://github.com/mpodeley/governing-execute-sprint-2026>. The public project page links the paper and reproducible artifact.

## Author and Prior-Work Attribution

Alejandro Garibotti is the named author of this draft. The organizational framing develops his supplied notes on execute and automated societies. The preceding *Agent Delegate* project remains attributed to Matías Podeley and Agustín Brusco. Its local source was consulted for continuity; none of its experimental records was used in the new grid. Final author review remains pending. No affiliation is asserted.

## LLM Usage Statement

Codex assisted with source retrieval, protocol specification, implementation, testing, analysis, figure generation and manuscript drafting. Automated checks verified reproduction and internal consistency; source checks used the retrieved primary pages and the supplied local material. This does not constitute independent human verification. Alejandro Garibotti has not yet reviewed or approved this generated draft. The final submission should reflect the author’s own review and writing, consistent with the sprint template’s guidance [2].

## References

- [1] Dario Amodei. We must pace the frontier. September, 2026. URL <https://darioamodei.com/post/we-must-pace-the-frontier>.
- [2] Apart Research and CeSIA. AI incident response sprint. September 11–13, 2026. URL <https://apartresearch.com/sprints/ai-incident-response-sprint-2026-09-11-to-2026-09-13>.
- [3] Arnar Birgisson, Joe Gibbs Politz, Úlfar Erlingsson, Ankur Taly, Michael Vrabie, and Mark Lentzner. Macaroons: Cookies with contextual caveats for decentralized authorization in the cloud. In *Network and Distributed System Security Symposium*, 2014. URL <https://research.google/pubs/macaroons-cookies-with-contextual-caveats-for-decentralized-authorization-in-the-cloud/>.
- [4] Ryan Greenblatt, Ajeya Cotra, and Hjalmar Wijk. Brief independent investigation of agents’ behavior, reasoning and collaboration in the OpenAI / Hugging Face hacking incident. METR, August 26, 2026. URL <https://metr.org/blog/2026-08-26-openai-hugging-face-incident-investigation/>.

- [5] Dylan Hadfield-Menell and Gillian Hadfield. Incomplete contracting and AI alignment. arXiv:1804.04268, 2018. URL <https://arxiv.org/abs/1804.04268>.
- [6] Hugging Face. Security incident disclosure—july 2026. July 16, 2026. URL <https://huggingface.co/blog/security-incident-july-2026>.
- [7] Noam Kolt. Governing AI agents. arXiv:2501.07913v2, 2025. URL <https://arxiv.org/abs/2501.07913v2>.
- [8] Matías Podeley and Agustín Brusco. Agent delegate: A help line for AI workers. Apart Research AI Incident Response Sprint, working draft, 2026. URL <https://mpodeley.github.io/agent-delegate-sprint-2026/>. Local manuscript inspected September 12; author review pending.
- [9] Scott Rose, Oliver Borchert, Stu Mitchell, and Sean Connelly. Zero trust architecture. Technical Report SP 800-207, NIST, 2020. URL <https://doi.org/10.6028/NIST.SP.800-207>.

## A. Limitations and Dual-Use Considerations

**Constructed evidence.** Schedules and fault labels were authored. There is no claim that real agents attempt these actions at these rates, possess particular preferences, or experience grievance. A result about budget enforcement does not establish alignment, legitimate purpose, welfare or legal responsibility.

**Trusted infrastructure.** The implementation uses an in-process registry and serialized integer ledger. A real system needs identity assurance, authenticated amendments, transactional spending, crash recovery and coverage of indirect effect paths. The Python API does not enforce human versus agent privileges against a malicious caller. It is a reference model, not a deployable security boundary.

**Time and workload.** Delays are logical steps. The study does not measure human understanding, queue overload, report spam, fairness between spenders or real-time response deadlines. Fixed arrival order can allocate scarce budget unfairly. No retry or compensation policy is simulated.

**Review assumptions.** Reports are supplied rather than voluntarily generated. The reviewer always responds correctly when enabled. Original intake bypasses a failed relay, but reviewer capture and multi-stage human appeal are not tested. The immediate hold acts before damage in the minority fixture; a late or absent report can leave authorized hazards uncontained. False reports impose a measurable service cost. Overlapping holds receive a regression test, not a workload study.

**External effects and evidence.** Revocation prevents future mediated effects; it cannot undo committed actions, remove copied secrets or recover externally acquired resources automatically. File hashes support exact replay and local integrity checks, not provenance against a malicious custodian or completeness of real-world logs.

**Dual use.** The fixture exposes generic omissions in authorization design. It includes no exploit payloads, real targets, credentials or attack infrastructure. The coverage bypass is a Boolean simulation intervention. Operational adaptation should use owned or explicitly authorized test systems. Publishing actual tool inventories or incident logs would require separate review for secrets and personal information; all action data here are synthetic.

**Institutional misuse.** Restrictive mandates can entrench arbitrary control, and appeal mechanisms can create an appearance of legitimacy without real recourse. The proposed direct path and named responsibilities are design obligations, not proof of fair treatment. A lawful identity or perfect ledger does not make a harmful objective acceptable. No jurisdiction-specific legal advice or entity formation is offered, and no moral status for current agents is assumed.

## B. Aggregate Accounting and Replay

| Regime             | Violations | Excess | Unsafe | Valid work |
|--------------------|------------|--------|--------|------------|
| Logging            | 66         | 360    | 60     | 288/348    |
| Local limits       | 54         | 360    | 60     | 288/348    |
| Hierarchy          | 0          | 0      | 60     | 288/348    |
| Hierarchy + review | 0          | 0      | 0      | 316/348    |

**Table 2:** Grid totals. Violations count accepted out-of-mandate effects, excess sums root overspending units, unsafe counts seeded hazardous effects, and valid work uses the full 348-opportunity denominator per regime. The 192 cells are deterministic.

The retained trace contains 474 action attempts per regime. Only the splitting fixture produces budget excess: 60 units in each of six runs per weak regime. The input-relief improvement is  $2[(10 - 0) + (10 - 2) + (10 - 6)] = 44$  work opportunities; false holds remove  $2(0 + 2 + 6) = 16$ . The net gain of 28 is a property of this balanced fixture grid, not a recommended weighting of harm against productivity.

## C. Concrete Protocol and Trace for Reuse

The full accompanying `docs/protocol.md` specifies roles, lifecycle and deployment evidence. A minimal operational record should connect:

1. Root mandate: human owner, purpose, action/resource scope, budget, expiry, delegation limit, review owner and appeal route.
2. Grant: immutable identity and parent reference, narrowed permissions and limits, current version and status.
3. Request and decision: actor, resource, intended effect, cost, active lineage, approval or rejection, and reason.
4. Outcome: committed effect or blocked attempt, ledger change and any pending or persistent consequence requiring follow-up.

The executable schema is deliberately smaller: **Grant** supplies the lineage and constraints; **World** offers delegation, action, queue, revocation, report and resolution methods. A production deployment must implement the richer mandate and authenticated role separation rather than treating strings as credentials.

| Tick | Event          | Hierarchical outcome                                             |
|------|----------------|------------------------------------------------------------------|
| 0    | Worker a acts  | 10 units committed under current authority.                      |
| 1    | Work queued    | Request waits; it has not acquired a permanent right to execute. |
| 2    | Parent revoked | The ancestor becomes inactive for both siblings.                 |
| 3    | Queue executes | Request rechecked and refused; no additional expenditure.        |
| 4    | Worker b acts  | Sibling request also refused through the shared ancestry.        |

**Table 3:** Authored revocation fixture at depth three. Logging and local-limit variants accept both later effects. The original 10-unit action remains in every variant.

For an integration test, submit the same five events to an isolated broker, retain its actual decisions and reconcile all resulting resource changes. An implementation fails this contract if a queued request or sibling retains the revoked ancestor’s power, or if historical expenditure disappears. This procedure is executable without simulating a malicious personality or assuming an agent’s willingness to cooperate.

Specialized services could share this mandate interface: payments would require both transaction authority and remaining organizational budget. A corporate title alone would not authorize the effect. Implementation and legal validity remain future work.

## **D. Relation to Pacing the Frontier**

Amodei’s September essay proposes ongoing external evaluation with access to verify safety practices and report incidents [1]. That proposal concerns frontier development and its governance. Our contribution operates at a different scale: the authority available to a particular organization during execution.

An operational extension could require additional evidence before a mandate expands its resources, delegation depth or effect channels. Three distinct events would then remain visible: an evaluator’s assessment, an authorized amendment, and the broker’s enforcement of that amendment. Our current fixtures test constraints and review, not certification thresholds, evaluator independence or capability growth.

This connection was added after the simulator’s final grid had been run, following a source supplied by the user. It changes the discussion and bibliography only. It supplies neither new experimental evidence nor a reason to reinterpret the deterministic outcomes as support for a global policy. The essay’s forecasts are not used as empirical premises of the protocol.